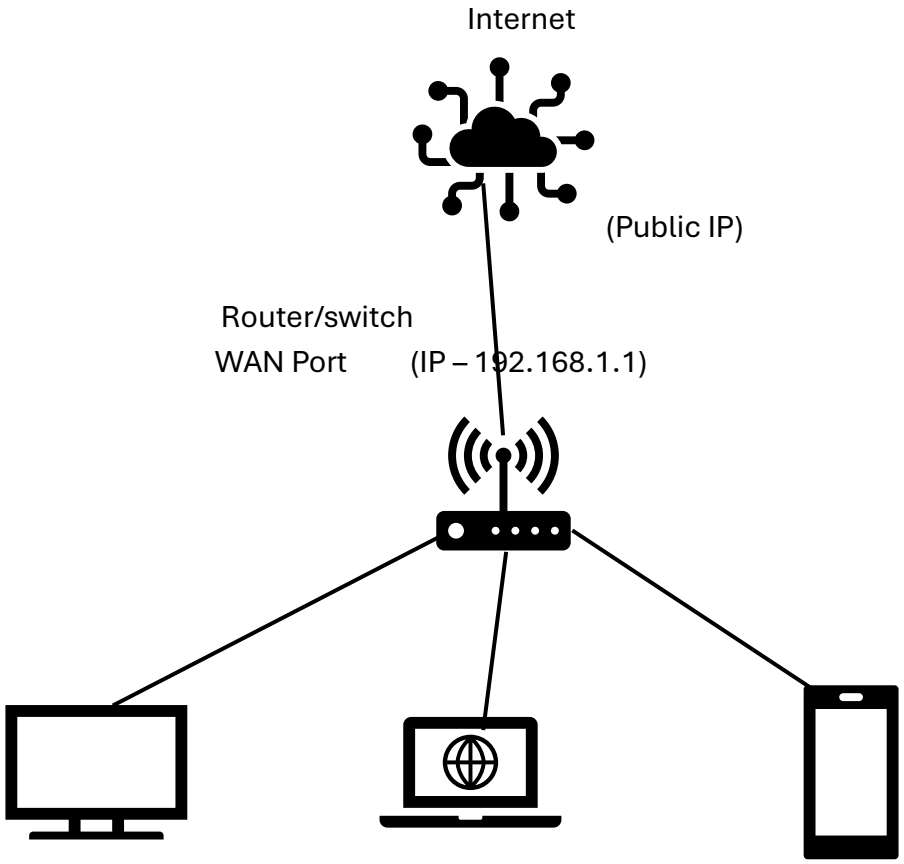
1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.
Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

The Setup

- Your Laptop is in Room A (192.168.10.5).
- Friend's Computer is in Room B (192.168.20.5).
- Router 1 guards Room A, and Router 2 guards Room B. They connect via a hallway network (10.0.0.x).

Step 1: Your Laptop Leaves Room A

Your laptop looks at the target address (192.168.20.5) and realizes it is not in its own local network (192.168.10.x).

- The Table Checked: Your laptop checks its local directory.

- The Decision: It sends the packet to its Default Gateway (Router 1 at 192.168.10.1) to handle it.

Step 2: Router 1 Hands it to Router 2

Router 1 receives the packet.

- The Table Checked: Router 1 looks at its internal road map:
    - *To get to 192.168.20.0, send it to Router 2 (10.0.0.2).*
- The Decision: Router 1 passes the packet down the hallway to Router 2.

Step 3: Router 2 Delivers the Packet

Router 2 receives the packet.

- The Table Checked: Router 2 looks at its road map:
    - *192.168.20.0 is directly connected to my other side!*
- The Decision: Router 2 spots your friend's computer right there in Room B and hands it the packet.

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Here is the routing table for a fictional **Zomato-style delivery network** hub, connecting three regional cloud kitchens/branches across India.

The Network Layout

- **Central Hub (Mumbai):** 10.100.0.0/16
- **Branch 1 (Delhi Kitchen):** Subnet 172.16.10.0/24
- **Branch 2 (Bangalore Kitchen):** Subnet 172.16.20.0/24
- **Branch 3 (Kolkata Kitchen):** Subnet 172.16.30.0/24

Central Mumbai Hub Router Routing Table

Destination Branch	Network Address	Subnet Mask	Next Hop (Gateway IP)	Outgoing Interface
Local Hub (Mumbai)	10.100.0.0	255.255.0.0	0.0.0.0 (Directly Connected)	gig0/0 (LAN)
Branch 1 (Delhi)	172.16.10.0	255.255.255.0	10.200.1.2 (Delhi Edge Router)	se0/0/0 (Leased Line)
Branch 2 (Bangalore)	172.16.20.0	255.255.255.0	10.200.2.2 (Bangalore Edge Router)	se0/0/1 (Leased Line)

Branch 3 (Kolkata)	172.16.30.0	255.255.255.0	10.200.3.2 (Kolkata Edge Router)	se0/1/0 (Leased Line)
Zomato App Cloud / Internet	0.0.0.0	0.0.0.0	192.0.2.1 (ISP Gateway)	gig0/1 (WAN)

How a Delivery Order Route Works

1. When a customer in **Delhi** places an order, the central Mumbai database sends the preparation ticket to 172.16.10.55 (a kitchen display screen in Delhi).
2. The Mumbai router scans this table, matches it to the **Delhi** row, and instantly forwards it out of interface se0/0/0 to the next hop (10.200.1.2).
3. The order travels instantly down that path straight to the Delhi kitchen.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Feature	**Static Routing (Manual Setup)**	**Dynamic Routing (Automatic - e.g., OSPF, RIP)**
Real-World Analogy	Pre-printed flyer with fixed, unchangeable driving directions.	Google Maps that instantly recalculates paths around traffic jams.
Zomato App Example	Connecting a local kitchen directly to its fixed local billing printer.	Rerouting order notifications through a backup line when the main internet drops.
Pro #1	Predictable: Traffic always takes the exact same path.	Self-healing: Instantly bypasses broken cables or hardware failures
Pro #2	Secure: No automated route sharing that hackers can exploit.	Scales easily: Automatically maps 100 new cloud kitchens when they open.
Pro #3	Lightweight: Uses zero router processor power or extra data bandwidth.	Traffic balancing: Splits heavy order traffic across multiple lines.
Con #1	No automatic backup: If a link dies, the network stays dead until manually fixed.	Resource heavy: Requires expensive routers with powerful processors.
Con #2	Hard to scale: Engineers must manually configure every single new connection.	Wastes bandwidth: Routers constantly chat with each other to share maps
Con #3	Human error: A single mistyped number can crash a city's network.	Complex fixes: Hard to troubleshoot if routes start glitching dynamically.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

Here is a standard, enterprise-grade configuration for setting up OSPF (Open Shortest Path First) routing between two Cisco routers, followed by an explanation of what each command does.

Router 1 (Mumbai Kitchen Hub) Configuration

(TEXT)

```
R1> enable
R1# configure terminal
R1(config)# interface gigabitethernet 0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
```

Router 2 (Delhi Kitchen Branch) Configuration

(TEXT)

```
R2> enable
R2# configure terminal
R2(config)# interface gigabitethernet 0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 0
```

Step-by-Step Command Breakdown

- **enable**
  What it does: Elevates access privileges. It is like logging into your computer as an "Administrator" so you are allowed to make deep system changes.

- **configure terminal**
  What it does: Opens the global configuration mode. Think of this as opening the main Settings panel on your phone where you go to tweak system options.

- **interface gigabitethernet 0/0**
  What it does: Zooms into the specific physical network port labeled 0/0. It tells the router, "The next changes I type apply only to the cable plugged into this specific slot."

- **ip address 10.0.0.1 255.255.255.252**
  What it does: Manually assigns an IP address and network boundary (subnet mask) to that physical slot so it can talk to the cable connected to Router 2.

- **no shutdown**
  What it does: Turns the interface port on. By default, Cisco ports are turned off for safety; this flips the power switch to "On" so electricity and data can flow.

- **router ospf 1**
  What it does: Fires up the OSPF dynamic routing engine. The number 1 is a local process label. It tells the router, "Start running the automatic map-sharing protocol right now."

- **router-id 1.1.1.1**
  What it does: Gives the router a unique name or ID tag. When the routers talk to each other to share maps, they use this ID to identify who is speaking (like a name badge).

- **network 10.0.0.0 0.0.0.3 area 0**
  What it does: Tells OSPF to activate on the link connecting Router 1 and Router 2. The 0.0.0.3 is a "wildcard mask" (an inverse subnet mask) that matches just those specific connecting IPs. area 0 is the core backbone zone where all routers must meet.

- **network 192.168.10.0 0.0.0.255 area 0**
  What it does: Tells OSPF to tell other routers about the local network attached to it (like the local Mumbai kitchen computers). It shouts to the network: *"Hey, if anyone wants to reach the Mumbai kitchen network, tell them to send it through me!"*

Once both routers are turned on with this config, they automatically send "Hello" packets to each other via their 10.0.0.x link, realize they speak the same OSPF language, merge their roadmaps, and instantly begin routing your Zomato order data dynamically.